

Mini Assessment

Name : Nallapureddy Mohan Reddy

Reg No : 99230040690

Dept : CSE

Date : 11-09-2026

Q1. If an API returns 429, what should your program do?

Answer: A 429 status code means “Too Many Requests” — the client has exceeded the API's rate limit.

- Check for a Retry-After header and wait for that duration before retrying.
- Use exponential backoff, which means waiting progressively longer between retries when no retry time is provided.
- Limit the number of retry attempts to avoid an infinite loop.
- Log the event and, if retries are exhausted, fail gracefully and show an appropriate error message to the user.

Q2. If response.json() fails, what should your program do?

Answer: response.json() can fail when the response body is empty, malformed, or is not valid JSON, such as an HTML error page.

- Handle the JSON parsing error using appropriate exception handling so the program does not crash.
- Check response.status_code and, when appropriate, response.text to understand what was actually returned.
- Avoid allowing the parsing error to terminate the whole program; return a safe error message or fallback value when suitable.
- Log useful information for debugging while avoiding exposure of sensitive response data.

Q3. If a JSON key is missing, why might .get() be safer than []?

Answer: Using square brackets, for example data['key'], raises a KeyError if the key does not exist.

- Using data.get('key') is safer because:
- It returns None, or another default value you specify, instead of raising an error.
- It allows the program to continue running even when JSON data is incomplete or inconsistent.
- A fallback value can be supplied, for example data.get('key', 'default_value').
- It makes the program more robust when handling real-world API responses.